

1. Problem definition:

1.1. What is the Problem?(note: possible names include - compass

Throughout their studies, students often lack clarity on what to pursue beyond compulsory coursework i.e. which electives, minors, extracurriculars, exchanges, internships, projects, etc. Consequently, many reach the end of their degree in a state of limbo, unable to determine how to progress further in their field, how to pivot to another, etc. often realising late that valuable opportunities had been available to them from the outset of their studies. Such choices directly shape a student's competitiveness for subsequent career paths, including postgraduate study or initial employment. Notably, the elements needed to construct such a path already exist, what is lacking is a means of integrating them into a coherent structure. Hence, relieving students of the burden of undertaking this synthesis unaided.

1.2. Subproblems

1.2.1. Outcome visibility gap

Students often see individual opportunities without really understanding what those choices will lead them to, which leads them to making choices without properly evaluating its long term outcome.

1.2.2. Lack of consolidation

Relevant opportunities are spread across university websites, courses, student associations, career portals, university databases, and informal sources.

1.2.3. Delayed feedback

Students discover whether their academic choices were useful much later. At that point it is usually too late and difficult to acquire the correct knowledge and skills.

1.2.4. Interest-to-choice gap

Students can have broad interests that they want to focus on, but they often struggle to put these interests into academic choices.

1.2.5. Changing goals and path adaptability

During studies, students frequently change their ambitions and interests. However, existing planning tools treat study planning as static rather than continuously reassessing what options remain available when a student changes direction.

2. Proposed solution

A study-path advisor sitting between a student's stated interests and the choices their programme still permits. Rather than presenting opportunities as a catalogue to browse, it works backwards from where the student wants to end up: it interprets a loosely described interest or destination, establishes what that destination demands, compares this against what the student has completed and can still fit into their remaining time, eliminates combinations the regulations make impossible, and returns a few coherent whole paths, each stating what it opens and closes off, re-evaluating whenever the student's direction changes. Interpreting destinations and opportunities is the part genuinely requiring intelligence: both are inconsistent free text written for different purposes, and no fixed rule maps one to the other. Feasibility, by contrast, follows institutional rules and requires no learning. The student remains the decision-maker; the system narrows and explains, but does not enrol, apply or decide, and presents no path as a guarantee.

Input from	What it provides	Output	Content
Student	Stated interest or destination in their own words; completed courses, programme, year, remaining credits; constraints (language, workload, exchange)	Ranked shortlist	Three to five coherent paths, each feasible in the remaining study time
Institution	Descriptions and learning outcomes of courses, minors, projects, internships, exchanges	Outcome statement	Which destinations each path opens — master's programmes, research areas, roles
Institution	Study regulations: prerequisites, credit rules, compulsory components, decision deadlines	Justification	What each element contributes, and why the paths are ordered as they are
Wider environment	Master's entry requirements; competencies named in graduate vacancy and internship listings	Trade-off warning	What each path makes harder or rules out
The system's own use	Which paths students accepted, rejected, or later moved away from	Timing signal	Which decisions must be made in which period, before the option expires

3. Goals/ Objectives:

3.1 Organisational objectives: Support students in choosing electives/minors and future master's paths aligned with their interest area.

3.2 Leading objectives: Structure electives, minors, and master's programs into interest-based paths and make path decisions navigable without needing an advisor.

3.3 User-oriented objectives: Students can identify their interest area, see which electives/minors fit that interest area, and then they can see which master's programs follow from those choices. Students should, as a result, have lower overall stress levels.

3.4 Component (model) objectives: Inventory of interest areas, inventory of electives/minors depending on the interest area, inventory of relevant master's programs. Ensure data privacy.

4. Measurements/KPIs

4.1 Percentage of system-recommended electives/minors that match manually checked sample of correct mappings

4.2 Average number of correctly tagged courses within the the top 5 recommendations per student

4.3 Uses a rating system(1-5) with feedback from students measuring stress levels and interest fit averaged across respondents.

4.4 How confident students feel about their choice after using the system (1–5) Target: Average 4 or higher.